13/10/2026 CO1-AT1-ITA0329 Name: M. Manoj Kumar

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1. $45\ MHz/(2 \times 25\ kHz)$= 900 channels

2. $560/7$= 80 channels per cell

3. 100×8= 800 voice channels

4. $10\log_{10}(80/4)$= 13.01 dB

5. $-60 - (-92)$= 32 dB

6. $(42\ MHz/300\ kHz)/7$= 20 channels per cell

7. $64/2^4$= 4 W

8. 2^2= 4 times

9. 4×15= 60 channels

10. $f_d = vf/c$= 83.33 Hz

11. $5/8$= 0.625 ms

12. $24\ MHz/200\ kHz$= 120 carriers

13. $10\log_{10}(8000) - 33$= 6.03 dBm

14. $FSPL = 32.44 + 20\log_{10}(1900) + 20\log_{10}(2.5)$= 105.97 dB

40. $43 - (-87)$= 130 dB

41. 150×2.5= 375 km²

42. $(48\ MHz/200\ kHz)/6$= 40 channels per cell

43. $150{,}000/6000$= 25 cycles

44. $9/3$= 3 handoffs (approximately)

45. $-85 - (-115)$= 30 dB

46. $250 \times 150\ kbps$= 37.5 Mbps

47. $1500 \times 4 \times (2.5/60)$= 250 Erlangs

48. $250 \times 12\ kHz$= 3 MHz

49. $1500 \times 600\ kbps$= 900 Mbps

50. $350/(\pi \times 4^2) = 6.96$= 7 towers